

VEX COMPETITIVE RATING

Full algorithm documentation

A complete numerical specification for ranking VEX V5 teams, events, regions, offensive and defensive strength, elimination performance, alliance contribution, and alliance-selection decision quality.

DEFINITIVE LAUNCH MODEL / VERSION 2.0

Prepared for the VEX V5 global ranking and statistics platform | September 2026

SYSTEM DEFINITION

1. Purpose and model philosophy

VEX Competitive Rating (VCR) estimates current competitive strength. It rewards performance above expectation, not attendance, raw score accumulation, geography, awards, or reputation. Every public change must be reproducible from frozen pre-event information and official results.

Primary outputs

Output	Scale	Meaning
Internal rating μ	Centered at 1500	Elo-style latent team strength
Public VCR	0-1000	Readable transformation of conservative strength
Rating deviation RD	40-350	Uncertainty; lower is more certain
Event Strength Score ESS	0-100	Field quality and competitive reliability
Event tier	C to Gold S / Worlds	Public event classification
Offense / Defense	Game-point units	Globally adjusted contribution estimates
ASE	0-100	Alliance Selection Efficiency; separate from VCR

Non-negotiable rules

- Judged awards never enter VCR, ESS, offense, defense, elimination, or regional-strength calculations.
- Raw OPR is never compared across different games and is never used alone for a global leaderboard.
- Event tier and team expectations are frozen before an event begins; future evidence cannot alter an earlier expectation.
- Country is never a permanent handicap. Weak or isolated evidence receives lower confidence until cross-region performance verifies it.
- All negative eligible results count. Positive-event limits prevent participation volume from becoming a ranking strategy.

Launch status

All constants marked launch default are initial values. They remain fixed for a published season unless a documented emergency correction is required. Future values are selected by chronological backtesting, not by manually matching popular opinion.

REQUIRED INPUTS

2. Data model and eligibility

The processor consumes official events, teams, matches, alliance assignments, scores, rankings, elimination brackets, autonomous outcomes, invitation/decline records when available, and geographical metadata.

Match eligibility

Condition	Treatment
Official scored match	Full weight
Tie	Actual result = 0.5
Team disqualification	Score result counts; reliability may be 0.75 if data cannot separate cause
No-show / disabled entire match	Result counts; MOV capped at 1.10; contribution reliability = 0.25
Replay supersedes original	Use final official replay only
Practice / skills / scrimmage	Excluded
Missing score	Excluded from score models; W/T/L may count at reliability 0.50
Correction after publication	Reprocess event and all later events chronologically

Event eligibility

Parameter	Launch default
Official tier minimum field	24 teams
Minimum official data completeness	95% of match outcomes
Tier freeze	24 hours before scheduled first match
Tier roster recalculation trigger	More than 10% of registered teams change after freeze
Provisional event	Fewer than 24 teams or below 95% completeness
Provisional event rating influence	Reliability capped at 0.70
Minimum offense leaderboard sample	12 valid matches and 2 events
Active-team window	At least one event in last 150 days

Canonical processing order

- Freeze entrant ratings and event tier.
- Process official matches in chronological order.
- Fit event contribution statistics after results are complete.
- Calculate qualification and elimination outcomes versus frozen expectations.
- Apply one event-close adjustment.
- Update uncertainty, recency evidence, regional links, and leaderboards.

CORE STATE

3. Rating state, initialization, and display

Each team-season stores latent rating μ , rating deviation RD, match count, event count, last-active date, and four component histories. Internal values use the Elo center; the site displays a bounded public scale.

Initialization

Team state	μ	RD
Completely new team	1500	350
Returning team after seasonal carryover	Formula below	Maximum 220
Established active team floor	Current	Minimum 40

$$\text{SeasonStartMu} = 1500 + 0.40 \times (\text{PreviousSeasonEndMu} - 1500)$$

$$\text{SeasonStartRD} = \min(220, \sqrt{\text{PreviousRD}^2 + 120^2})$$

The 40% carryover recognizes persistent program quality while allowing a new game, robot, roster, and strategy to redefine strength.

Conservative strength and public VCR

$$\text{ConservativeMu} = \mu - 0.75 \times \text{RD}$$

$$\text{PublicVCR} = \text{clamp}(0, 1000, 500 + 1.25 \times (\text{ConservativeMu} - 1500))$$

Example: μ 1700 and RD 60 gives ConservativeMu 1655 and PublicVCR 693.75, displayed as **694**. Global ordering uses unrounded ConservativeMu, so displayed ties remain deterministic.

Tie-break sequence

Order	Tie-break
1	Higher unrounded ConservativeMu
2	Lower RD
3	Higher last-90-day event evidence
4	Higher elimination component
5	Team number ascending

MATCH ENGINE

4. Opponent-adjusted match model

For red alliance A+B against blue alliance C+D:

$$R_{red} = (\mu_A + \mu_B)/2 \quad R_{blue} = (\mu_C + \mu_D)/2$$

$$E_{red} = 1 / (1 + 10^{((R_{blue} - R_{red})/400)}) \quad E_{blue} = 1 - E_{red}$$

Actual is 1.0 for a win, 0.5 for a tie, and 0.0 for a loss. Base surprise equals Actual minus Expected.

Margin of victory

$$MOV = 1 + 0.35 \times \tanh(\text{ScoreMargin} / \max(\text{EventScoreSD}, 12))$$

Launch Kmatch = 20. Qualification MOV is capped at 1.30; elimination MOV at 1.20 because advancement is separately rewarded. A no-show match is capped at 1.10.

$$\text{AllianceDelta} = K_{\text{match}} \times (\text{Actual} - \text{Expected}) \times MOV \times \text{MatchReliability}$$

Match reliability condition	Value
Normal complete official match	1.00
Partial scoring detail but valid outcome	0.85
DQ with ambiguous individual cause	0.75
Outcome only / missing score	0.50
Major no-show or disabled-data ambiguity	0.25 for contribution; 1.00 for W/T/L

Do not multiply ordinary match updates by Event Weight. The field's ratings already set match difficulty. Event Weight is applied to event-close evidence and elimination surprise, preventing double-counting of opponent strength.

CREDIT ALLOCATION

5. Dynamic alliance contribution sharing

The fixed 35/65 split is replaced by a dynamic 20/80 range. Current-event evidence outweighs global rating once enough matches exist. A team is never granted credit solely because it was already highly ranked.

Evidence weights by matches completed before the evaluated match

Prior valid matches	Global VCR	Event performance	Event rank
0-1	60%	35%	5%
2-3	40%	50%	10%
4-5	30%	56%	14%
6+	20%	60%	20%

Global confidence multiplier

Prior event history	Global confidence
0 events	0.10
1 event	0.35
2 events	0.60
3+ recent events	1.00

Multiply the global weight by confidence and transfer unused weight proportionally to Event Performance and Event Rank. Standardize all three inputs before combination.

$$\text{LocalStrength}_i = \text{wg } Z(\text{Global}_i) + \text{wp } Z(\text{EventPerformance}_i) + \text{wr } Z(\text{EventRank}_i)$$

$$\text{RawShareA} = \exp(\text{LocalStrengthA}/1.5) / [\exp(\text{LocalStrengthA}/1.5) + \exp(\text{LocalStrengthB}/1.5)]$$

$$\text{ShareA} = \text{clamp}(\text{DynamicMin}, \text{RawShareA}, \text{DynamicMax}); \text{ShareB} = 1 - \text{ShareA}$$

Absolute partner VCR gap	Stronger-team maximum	Weaker-team minimum
0-99	0.55	0.45
100-199	0.65	0.35
200-299	0.725	0.275
300+	0.80	0.20

For match N, use event evidence from matches 1 through N-1. At event close, a leave-one-match-out recalculation may evaluate each match using every other match, preventing a match from justifying its own credit allocation.

Loss responsibility

$$\text{LossShare}_i = 0.50 \times \text{ExpectedContributionShare}_i + 0.50 \times \text{UnderperformanceShare}_i$$

This prevents a strong team from assigning every loss to a weaker partner. Underperformance is the positive magnitude of the team's contribution residual below expectation, normalized across both partners.

CONTRIBUTION MODEL

6. Global offense, defense, OPR, DPR, and CCWM

Raw event OPR is descriptive, not a global strength measure. Global estimates must adjust for partners, opponent defense, event environment, limited schedules, and disconnected regions.

Regularized score model

$$\text{AllianceScore} = \text{GameBaseline} + \text{OffA} + \text{OffB} - \text{DefC} - \text{DefD} + \text{EventEnvironment} + \text{Error}$$

Fit offense and defense with ridge regression. Standardize design-matrix columns. Launch ridge penalty $\lambda = 8.0$, selected later by five-fold event-grouped cross-validation over $\{1, 2, 4, 8, 16, 32\}$. Never split matches from the same event across training and validation folds.

Derived values

Metric	Definition	Public direction
Adjusted OPR	Estimated offensive point contribution	Higher is better
Adjusted DPR	Estimated points prevented	Higher is better in VCR storage
CCWM	Adjusted OPR + Adjusted DPR	Higher is better
Offense conservative score	Adjusted OPR - $1.0 \times \text{SE(OPR)}$	Higher is better
Defense conservative score	Adjusted DPR - $1.0 \times \text{SE(DPR)}$	Higher is better

Because conventional DPR definitions sometimes use points allowed, ingestion must convert them to the internal 'points prevented' direction once. Never negate the value again downstream.

Contribution composite

$$\text{ContributionZ} = 0.55 \text{ Z(AdjOPR)} + 0.20 \text{ Z(AdjDPR)} + 0.25 \text{ Z(CCWM_residual)}$$

CCWM_residual is the portion of CCWM not already explained by adjusted OPR and DPR. Residualization prevents the identity CCWM approximately equals OPR minus conventional DPR from being counted repeatedly.

Event performance used for partner sharing

$$\text{EventPerformance} = 0.45 \text{ Z(AdjOPR)} + 0.20 \text{ Z(AdjDPR)} + 0.20 \text{ Z(CCWM_residual)} + 0.15 \text{ Z(Auto)}$$

GAME-PHASE MODEL

7. Autonomous performance

Autonomous is treated as both an alliance outcome and an estimated individual contribution. The Autonomous Win Point (AWP) is never credited equally without adjustment because partners and opponents affect the probability.

$$PAWP = \text{logistic}(b0 + \text{AutoA} + \text{AutoB} - \text{AutoC} - \text{AutoD})$$

$$AWPResidual = \text{ActualAWP} - PAWP$$

$$\text{AutoScore} = 0.60 Z(AWPResidualRate) + 0.40 Z(\text{AutoContribution})$$

If individual autonomous scoring detail is unavailable, set AutoScore = Z(AWPResidualRate), reduce AutoReliability to 0.65, and redistribute the missing 40% internally rather than treating it as zero.

Autonomous constants

Parameter	Launch default
Minimum AWP sample for display	6 eligible qualification matches
AWP logistic ridge penalty	8.0
Auto contribution weight when detailed data exists	40%
AWP residual weight	60%
Missing-detail reliability	0.65
Autonomous share of event-close VCR update	10%

Game-specific rule configuration

The definition of eligible autonomous outcomes, ties, bonus completion, and point attribution belongs in a season rules file. Algorithm code must not hard-code one year's field elements. When a season has no AWP, replace AWPResidualRate with autonomous-period win probability residual.

TOURNAMENT OUTCOMES

8. Qualification and elimination performance

Tournament advancement is materially important. Event Result receives 17% of the event-close update, and elimination receives 65% of Event Result. Expected performance prevents a famous team from gaining simply for meeting expectations.

$$\text{ResultPercentile} = 0.35 \times \text{QualificationPercentile} + 0.65 \times \text{EliminationPercentile}$$

$$\text{EventSurprise} = \text{ResultPercentile} - \text{ExpectedResultPercentile}$$

Outcome	Elimination percentile
Champion	1.00
Finalist	0.92
Semifinalist	0.82
Quarterfinalist	0.70
Round of 16	0.55
First elimination round	0.40
Qualified but not selected	0.25
Did not qualify	$0.05 + 0.15 \times \text{QualificationPercentile}$

For bracket formats with different rounds, map advancement continuously: losing-round percentile = $0.25 + 0.65 \times$ fraction of elimination rounds survived, capped below finalist. Published event adapters must document the mapping.

Elimination importance

Tier	Elimination modifier	Elimination component cap
C	0.85	+/-25
B	0.90	+/-25
A	1.00	+/-35
Bronze S	1.10	+/-35
Silver S	1.20	+/-45
Gold S	1.30	+/-45
Worlds	1.45	+/-60

$$\text{DeltaElim} = \text{clamp}(\text{cap}, 55 \times \text{EliminationSurprise} \times \text{EventWeight} \times \text{EliminationModifier})$$

A team expected at 0.72 that records 0.15 at Worlds has surprise -0.57. Raw change is approximately -84 and is capped at -60. Missing eliminations therefore matters, but cannot erase an exceptional season in one step.

Alliance-selection protection

A strong unselected team is not treated identically to a team that performs poorly in eliminations. Qualification percentile, contribution residuals, selection status, DQs, substitutions, and reliability soften or redistribute the outcome. The bracket result remains real evidence, not an absolute measure of one robot.

EVENT RANKING

9. Event Strength Score and tiers

ESS uses frozen pre-event team information. Event branding sets eligibility ceilings, not automatic status.

$$\text{ESS} = 0.30 \text{ FieldStrength} + 0.25 \text{ FieldDepth} + 0.15 \text{ EliteDensity} + 0.15 \text{ RegionalCalibration} + 0.10 \text{ FormatQuality} + 0.05 \text{ RegionalDiversity}$$

Component	Weight	Exact launch definition
Field Strength	30%	Percentile of trimmed mean entrant ConservativeMu; trim top/bottom 5%
Field Depth	25%	60% top-quarter mean percentile + 40% top-half mean percentile
Elite Density	15%	50% top-1% density + 30% top-5% + 20% top-10%, percentile normalized
Regional Calibration	15%	Entrant-weighted Regional Strength Index
Format Quality	10%	See numerical table below
Regional Diversity	5%	Effective-region diversity normalized against same event class

$$\text{Format Quality} = 0.35 \text{ MatchCount} + 0.25 \text{ FieldSize} + 0.25 \text{ DataCompleteness} + 0.15 \text{ BracketIntegrity}$$

Subscore	0 points	50 points	100 points
Qualification matches/team	4 or fewer	6	8 or more
Field size	12 or fewer	24	48 or more
Data completeness	Below 80%	90%	100%
Bracket integrity	Invalid/missing	Reduced bracket	Complete standard bracket

Interpolate linearly between anchors and clamp each subscore to 0-100.

Tier thresholds

Tier	ESS	Event Weight	Eligibility / typical event
C	0-34	0.70	Any official local event
B	35-49	0.82	Strong local / average regional
A	50-64	0.95	Strong regional competition
Bronze S	65-74	1.15	Regional championship or Signature
Silver S	75-87	1.35	Signature only
Gold S	88-100	1.60	Signature only
Worlds	Special	1.85	Official World Championship

Ordinary regional events cannot exceed Bronze S. Regional championships can reach Bronze S. Signature Events can be Bronze, Silver, or Gold S; a Signature label does not guarantee strength.

UPSIDE AND DOWNSIDE

10. Event Weight and asymmetric expectation

Event Weight changes the value of event-close surprise. High-tier overperformance earns more; a favorite failing at a weak event is punished more. This requires an asymmetric rule.

$$\text{ContextWeight} = \text{EventWeight when Surprise} \geq 0; 1/\text{EventWeight when Surprise} < 0$$
$$\text{DeltaEventBase} = 80 \times \text{EventSurprise} \times \text{ContextWeight} \times \text{EventReliability}$$

DeltaEventBase is capped at +/-80 before component blending. Elimination uses its separate formula and cap from Section 8.

Worked values before K and reliability

Surprise	C 0.70	A 0.95	Bronze S 1.15	Silver S 1.35	Gold S 1.60
+0.20	+0.140	+0.190	+0.230	+0.270	+0.320
-0.20	-0.286	-0.211	-0.174	-0.148	-0.125

Event reliability

$$\text{EventReliability} = 0.40 \text{ DataCompleteness} + 0.30 \text{ MatchSample} + 0.20 \text{ ModelConnectivity} + 0.10 \text{ BracketIntegrity}$$

Input	Launch mapping
DataCompleteness	Observed valid matches / expected matches
MatchSample	min(1, average valid matches per team / 8)
ModelConnectivity	min(1, connected cross-event opponents / 12), floor 0.50
BracketIntegrity	1.00 complete; 0.75 reduced; 0.50 missing

Event Weight never excuses weak match results. It controls how much a tournament-level result changes confidence in a team relative to the field.

GLOBAL FAIRNESS

11. Regional ranking and calibration

Each team is shown in global, continental, country/region, global offense, and regional offense tables. Regional tables filter the same underlying global model; they do not create inflated parallel ratings.

Regional Strength Index

$$\text{RSI} = 0.35 \text{ CrossRegionResults} + 0.25 \text{ TopTeamStrength} + 0.20 \text{ RegionalDepth} + 0.10 \text{ InternationalParticipation} + 0.10 \text{ DataReliability}$$

Factor	Exact definition	
CrossRegionResults	Percentile of exponentially weighted Actual-minus-Expected result in cross-region matches	
TopTeamStrength	Trimmed mean ConservativeMu of top 10-20% active teams; minimum 5 teams	
RegionalDepth	50% regional median + 50% upper-half mean, percentile normalized	
InternationalParticipation	Effective number of cross-region opponents, saturating at 40	
DataReliability	min(1, valid regional matches/300) x min(1, cross-region links/40)	
RSI state	Regional factor	Use
Unknown / isolated	0.85	High uncertainty; not a country penalty
0-34	0.80-0.90	Weak evidence environment
35-49	0.90-0.97	Below average
50-64	0.97-1.03	Average
65-79	1.03-1.08	Strong
80-100	1.08-1.12	Elite/deep

The factor is used inside RegionalCalibration in ESS; do not multiply it onto Event Weight again.

Isolated-region shrinkage

$$\text{RegionalConfidence} = \text{CrossRegionEvidence} / (\text{CrossRegionEvidence} + 12)$$

$$\text{CalibratedMetric} = \text{Confidence} \times \text{Estimate} + (1 - \text{Confidence}) \times \text{GlobalMean}$$

Cross-region evidence	Confidence
0	0%
3	20%
6	33%
12	50%
24	67%
48	80%

Repeated local opposition

$$\text{NoveltyWeight} = 1 / \sqrt{1 + \text{RepeatedOpponentCount}}$$

This weight affects new evidence and uncertainty reduction, not whether the result counts. First exposure is 100%, second 71%, fourth 50%, ninth 33%, and sixteenth 25%.

ONE-YEAR SEASON

12. Season phases, recency, and inactivity

Past evidence decays; the underlying rating is not automatically destroyed. This avoids inactivity punishment while ensuring one early Signature victory cannot control the late-season leaderboard.

$$\text{RecencyWeight}(\text{days}) = 2^{(-\text{days}/75)}$$

Age	Remaining influence
0 days	100%
30 days	75.8%
75 days	50%
150 days	25%
225 days	12.5%
300 days	6.25%
365 days	3.4%

Season phases

Phase	Calendar portion	Current-season evidence	Starting prior
Early	First 30%	60%	40%
Mid	Next 35%	80%	20%
Late	Final 35%	95%	5%

$$\text{EvidenceWeight}_i = \text{EventWeight}_i \times \text{RecencyWeight}_i \times \text{Reliability}_i$$

$$\text{RecentPerformance} = \frac{\sum(\text{EvidenceWeight}_i \times \text{Performance}_i)}{\sum(\text{EvidenceWeight}_i)}$$

Single-event dominance controls

Control	Launch default
Maximum one-event share of positive evidence	30% after team has 3+ events
Confirmation multiplier: only one positive event	0.75
Two consistent events	0.90
Three or more consistent events	1.00
Low-tier positive-event count	Best 2 C/B bonuses in rolling 90 days
Negative events	All count

Inactivity

$$\text{After a 30-day grace period: } \text{RD}_{\text{new}} = \min(350, \text{RD}_{\text{old}} + 0.20 \times \text{InactiveDaysBeyondGrace})$$

This lowers conservative rank through uncertainty without claiming the team definitely became worse.

DECISION QUALITY

13. Alliance Selection Efficiency

ASE is a separate 0-100 profile metric. It measures whether an alliance captain selected a strong available partner using information available at selection time. ASE never enters VCR.

$$\text{PickValue} = 0.35 \text{ AllianceStrength} + 0.20 \text{ AutoCompatibility} + 0.15 \text{ StrategicCompatibility} + 0.15 \text{ Reliability} + 0.10 \text{ ScheduleAdjustedPerformance} + 0.05 \text{ SelectionFlexibility}$$
$$\text{OptimalValue} = \max \text{ PickValue among genuinely available teams}$$
$$\text{PickRegret} = \text{OptimalValue} - \text{SelectedPickValue}$$
$$\text{PickEfficiency} = \text{clamp}(0, 100, 100 \times [1 - \text{PickRegret}/\text{MaximumReasonableRegret}])$$

ASE	Interpretation
90-100	Near-optimal selector
75-89	Strong selector
60-74	Generally reasonable
40-59	Questionable / average uncertainty
0-39	Repeatedly poor choices

Selection weights

Decision	Importance
First partner	1.00
Second partner	0.70
Third / substitute	0.40

$$\text{HistoricalASE} = \text{sum}(\text{Recency} \times \text{EventWeight} \times \text{Importance} \times \text{Efficiency}) / \text{sum}(\text{weights})$$
$$\text{PublishedASE} = [n/(n+5)] \times \text{ObservedASE} + [5/(n+5)] \times 50$$
$$\text{Previous-season carryover: StartingASE} = 50 + 0.40 \times (\text{PreviousASE} - 50)$$

Reconstruct prior selections, declines, captain availability, pick order, and event rules. Never penalize a team for not selecting an unavailable or declining team. If compatibility data is missing, show low confidence and rely on projected alliance strength.

FINAL TEAM CALCULATION

14. Complete event-close update

After all event models are available, calculate four non-overlapping component deltas. The launch blend gives elimination and tournament outcome materially more influence than the original prototype.

$$\text{DeltaVCR_internal} = 0.55 \text{ DeltaMatch} + 0.18 \text{ DeltaContribution} + 0.10 \text{ DeltaAuto} + 0.17 \text{ DeltaEventResult}$$

Component	Weight	Component range before blend
Match	55%	Natural match sum; event-normalized and capped +/-100
Contribution	18%	+/-80
Autonomous	10%	+/-60
Event Result	17%	+/-80; elimination subcomponent has tier caps

$$\text{FinalEventDelta} = \text{clamp}(-100, +100, \text{DeltaVCR_internal})$$

$$\text{mu_after} = \text{mu_before} + \text{FinalEventDelta}$$

RD reduction

$$\text{EffectiveMatches} = \text{sum}(\text{MatchReliability} \times \text{NoveltyWeight})$$

$$\text{RD_after} = \text{max}(40, \text{RD_before} / \text{sqrt}(1 + \text{EffectiveMatches}/6))$$

An established team therefore stabilizes gradually; a new team can move quickly but remains conservatively ranked until evidence accumulates.

Double-counting controls

- Match outcomes carry most weight; contribution uses residualized statistics.
- Qualification and elimination MOV caps are lower because advancement is separately modeled.
- Event Weight does not multiply ordinary Elo match updates.
- Regional calibration is inside ESS and is never applied a second time.
- CCWM is residualized after adjusted offense and defense.

NUMERICAL WALKTHROUGH

15. Worked end-to-end example

Team X enters a Silver S event at mu 1710, RD 70. Its conservative strength is 1657.5, giving public VCR 697. The event has ESS 81, Event Weight 1.35, and reliability 0.94.

Match evidence

Across eight matches, opponent-adjusted alliance deltas allocated through dynamic partner shares sum to +34. MOV and reliability are already included. DeltaMatch = +34.

Contribution and autonomous

Adjusted OPR, defense, and residual CCWM produce ContributionZ +0.80, mapped to DeltaContribution +28. AutoScore +0.45 maps to DeltaAuto +14.

Tournament result

Qualification percentile = 0.78. The team reaches semifinals, elimination percentile 0.82. ResultPercentile = $0.35(0.78) + 0.65(0.82) = 0.806$. Frozen expected percentile was 0.65, so EventSurprise = +0.156.

$$\text{Base event surprise} = 80 \times 0.156 \times 1.35 \times 0.94 = +15.84$$

Elimination surprise subcomponent remains within the Silver S +/-45 cap. The combined Event Result delta is +22 after qualification/elimination decomposition.

Final blend

$$\text{Delta} = 0.55(34) + 0.18(28) + 0.10(14) + 0.17(22) = 28.88$$

Round internal event delta to four decimals, not an integer. New mu = 1738.88. If RD falls from 70 to 48, ConservativeMu = 1702.88 and public VCR = 754 after rounding.

Interpretation

The team improves because it beat expectations across matches, contribution, autonomous, and advancement. Its public VCR rises more than mu alone suggests because the event also reduced uncertainty. The site must show both effects separately.

PRODUCT RULES

16. Leaderboards and publication

Global and regional pages must make uncertainty, recency, and change explanations visible. Avoid presenting model estimates as exact truths.

Required team leaderboard fields

Field	Example
Global rank	#12
Regional rank	Europe #3 / Country #1
Public VCR	754
Internal mu	1738.88
Confidence	RD 48 - High
Recent change	+29
Last event	Silver S (1.35)
Change explanation	+18.7 match, +5.0 contribution, +1.4 auto, +3.7 result
Activity	8 days ago

Confidence labels

RD	Label
40-69	High
70-119	Medium
120-199	Low
200-350	Provisional

Required event page fields

- Tier, ESS, Event Weight, reliability, freeze timestamp, and roster snapshot.
- Each ESS component and its numeric score.
- Expected versus actual team percentile and VCR change decomposition.
- Tier eligibility ceiling and reason for any provisional status.

Required offense leaderboard fields

- Adjusted OPR, standard error, conservative offense, sample size, region confidence, last event, and game season.
- Raw event OPR may be shown for context but must be clearly labeled non-global.

MODEL VALIDATION

17. Backtesting, calibration, and governance

All changes are evaluated chronologically. A model may train only on information available before the predicted match or event.

Primary validation metrics

Metric	Target / use
Match log loss	Primary optimization metric; lower is better
Brier score	Probability accuracy
Calibration error	Predicted 70% outcomes should occur about 70%
Rank correlation	Future event performance versus current rating
Regional calibration gap	Maximum 3 percentage-point win-probability error by region
Tier calibration gap	Maximum 4 percentage-point error by event tier

Baselines

- Win percentage only
- Plain Elo with K=20
- Raw OPR only
- Elo plus adjusted OPR
- Current VCR without Event Weight
- Current VCR without region shrinkage

Parameter search ranges

Parameter	Launch	Search range
Elo scale	400	300-500
Kmatch	20	12-32
Evidence half-life	75 days	45-120
Event K	80	50-110
Elimination K	55	35-75
Conservative RD multiplier	0.75	0.5-1.25
Ridge lambda	8	1-32 log grid
Partner softmax temperature	1.5	0.8-2.5
Single-event positive cap	30%	20-40%

Governance

Freeze model version, constants, and code hash for the competitive season. Publish change notes. Do not silently revise history. If an official score correction occurs, label the recalculation as data correction rather than model change. Maintain separate development and production leaderboards.

REFERENCE PIPELINE

18. Implementation pseudocode

The following order is normative; implementation language is not.

1. load season configuration and prior-season carryover
 2. initialize mu, RD, region, and component histories
 3. for each event in start-time order:
 - a. freeze entrants, mu, RD, RSI, ESS, tier, and Event Weight
 - b. for each official match in time order:
 - i. compute alliance expectations
 - ii. compute MOV and reliability
 - iii. compute dynamic partner shares from prior local evidence
 - iv. store match component deltas
 - c. fit ridge offense, defense, autonomous, and residual models
 - d. optionally recompute shares leave-one-match-out
 - e. calculate qualification and elimination percentiles
 - f. calculate event surprise and asymmetric Event Weight
 - g. blend four component deltas and cap at +/-100
 - h. update mu and RD
 - i. append immutable audit record
4. apply recency evidence weights at leaderboard query time
5. compute ConservativeMu and PublicVCR
6. publish global, regional, offense, defense, event, and ASE views

Numerical precision

Value	Storage / rounding
mu, RD, probabilities, deltas	64-bit float; retain at least 6 decimals
Public VCR	Nearest integer, ordering by unrounded value
ESS / RSI / ASE	One decimal displayed; full precision stored
Event Weight	Two decimals displayed and stored from versioned table
Timestamps	UTC ISO 8601
Randomness	No randomness in production calculation

ALL LAUNCH NUMBERS

19. Constants registry

Constant	Value	Section
New-team mu	1500	3
New-team RD	350	3
RD floor	40	3
Returning RD cap	220	3
Season carryover	40%	3
Conservative RD multiplier	0.75	3
Public scale multiplier	1.25	3
Elo denominator scale	400	4
Kmatch	20	4
MOV coefficient	0.35	4
Score-SD floor	12	4
Partner softmax temperature	1.5	5
Partner absolute bounds	20%-80%	5
Ridge lambda	8.0	6/7
Auto weights	60% residual / 40% contribution	7
Qualification / elimination	35% / 65%	8
Kelim	55	8
Event K	80	10
Event-close cap	+/-100	14
ESS weights	30/25/15/15/10/5	9
Event weights	0.70/0.82/0.95/1.15/1.35/1.60/1.85	9
Region confidence denominator	12	11
Recency half-life	75 days	12
Inactivity grace	30 days	12
RD inactivity growth	0.20/day	12
Single positive event cap	30%	12
ASE prior strength	5 decisions at score 50	13
Final blend	55%/18%/10%/17%	14
RD effective-match divisor	6	14

This registry is the authoritative launch-default summary. Detailed definitions and caps in the cited sections control if a shorthand entry appears ambiguous.

OPERATIONAL READINESS

20. Known limitations and launch checklist

Known limitations

- Alliance-based scoring cannot perfectly identify individual contribution without richer telemetry.
- Disconnected regions remain uncertain until cross-region events create statistical anchors.
- Alliance selection contains private scouting and compatibility information the public model may not observe.
- Game-specific scoring changes require season-specific configuration and revalidation.
- Elimination advancement contains selection and bracket luck; caps and expectation adjustment reduce but do not remove it.

Launch checklist

Check	Required status
Official data adapters tested	Pass
Season rules file versioned	Pass
Chronological replay deterministic	Pass
No future-data leakage tests	Pass
Regional and tier calibration reported	Pass
Rating-change decomposition sums exactly	Pass
Score corrections reproducible	Pass
Public methodology page matches production constants	Pass
Baseline comparison published	Pass
Model version and code hash displayed	Pass

Recommended launch name

VCR 2.0 - VEX Competitive Rating
A predictive, opponent-adjusted, event-weighted global ranking system with regional confidence and transparent tournament-performance accounting.

End of specification.